

CHUNYIN, SIU (ALEX)

Brain Dynamics Lab, 1520 Page Mill Rd, Palo Alto, CA 94304

siuc@stanford.edu $\diamond$ <https://c-siu.github.io>

EDUCATION

- Cornell University**, Ithaca, NY *2019 – 2024*
PhD Applied Mathematics; supervised by Prof Gennady Samorodnitsky
- The Chinese University of Hong Kong (CUHK)**, Hong Kong *2017 – 2019*
MPhil Mathematics; supervised by Prof Ronald Lui
- The Chinese University of Hong Kong (CUHK)**, Hong Kong *2013 – 2017*
BSc Mathematics; Minor in Economics

EMPLOYMENT

- Postdoctoral Scholar, Stanford University School of Medicine** *2024 – present*
develop topological-statistical techniques to analyze neuroimaging data and identify behavioral correlates
- Affiliate, Lawrence Berkeley National Laboratory** *Summer 2023*
built a neural network to predict the adsorption loadings of zeolite crystals with their topological features
verify a conjecture on the universality of a topological statistic of scientific datasets.

ONGOING WORKS

Superscripts indicate career stages (UnderGraduate, Post-Bacc, Graduate Student, Post-Doc, Professor).

- C. Siu^{PD}, S. Madsen^{PB}, S. Quah^{PD}, C. Glick^{PB}, M. Saggarr^P. Revealing the Network Organization of Evoked Brain Activity.
- C. Siu^{PD}. The Topology of the Energy Landscape of Positive Orthogonally Decomposable Tensors

PREPRINTS AND PUBLICATIONS

$\diamond$ *indicates alphabetically-sorted author list. Superscripts indicate career stages as in the previous section.*

- C. Siu^{PD}, S. Pirzada^{PB}, C. Glick^{PB}, R. Betzel^P, G. Petri^P, L. Williams^P, M. Saggarr^P. Global Topology of Brain-wide Co-fluctuations Links Task States, Personality, and Behavioral Symptom Dimensions. Preprint.
- C. Siu^{GS}. "The Topological Behavior of Preferential Attachment Graphs". *SIAM Journal on Applied Algebra and Geometry*, 2025.
- C. Siu^{GS}, G. Samorodnitsky^P, C. Yu^P, and R. He^{UG}. "The Asymptotics of the Expected Betti Numbers of Preferential Attachment Clique Complexes". *Advances in Applied Probability*, 2025.
- C. Siu^{GS}, G. Samorodnitsky^P, C. Yu^P, and A. Yao^{UG}. "Detection of Small Holes by the Scale-Invariant Robust Density-Aware Distance (RDAD) Filtration". *Journal of Applied and Computational Topology*, 2024.
- $\diamond$ C. Siu^{GS}, and R. Strichartz^P. "Geometry and Laplacian on Discrete Magic Carpets". *Journal of Fractal Geometry*, 2023.
- H. Law^{GS}, C. Siu^{GS}, and R. Lui^P. "Decomposition of Longitudinal Deformations via Beltrami Descriptors". *Journal of Scientific Computing*, 2021.
- C. Siu^{GS}, H.L. Chan^{GS}, and R. Lui^P. "Image Segmentation with Partial Convexity Shape Prior Using Discrete Conformality Structures". *SIAM Journal on Imaging Sciences*, 2020.
- $\diamond$ J. Li^{UG}, and C. Siu^{UG}. "An Elementary Approach on Left-Orderability, Cables of Torus Knots and Dehn Surgery". Preprint.

TALKS AND POSTER PRESENTATIONS

* *invited talks* • *contributed talk* ◇ *poster presentation*.

"Topological Data Analysis – Connecting Topology, Probability, and Neuroscience".

- * Topology Seminar. Stanford University, CA, Oct 2025.

"A Novel Topological Characterization of Brain-wide Cofluctuation Patterns over Time Reveals Brain-Behavior Links across Spontaneous and Evoked Activity".

- Computation Persistence Workshop. Albany University, NY, Oct 2025.

"The Topology of Preferential Attachment Clique Complexes – Homology and Homotopy".

- SIAM Applied Algebraic Geometry Conference, University of Wisconsin-Madison, WI, Jul 2025.
- * Applied Algebraic Topology Research Network (AATRN) Networks Seminar. Virtual, Feb 2025.
- ◇ Mid-Atlantic Topology Conference, Northeastern University, MA, Mar 2024.
- * University of Florida Topological Data Analysis Conference. University of Florida, FL, Feb 2024.
- Northeast Probability Seminar, New York University, NY, Nov 2023.

"The Topology of Preferential Attachment Clique Complexes – Homology".

- * Probability Seminar. Stanford University, CA, Jun 2025.
- * Probability Seminar. Northwestern University, IL, Feb 2025.
- * Mathematics Seminar. The Chinese University of Hong Kong, Hong Kong, Dec 2023.
- Northeast Probability Seminar, New York University, NY, Nov 2023.
- Binghamton University Graduate Combinatorics, Algebra and Topology. Binghamton University, NY, Nov 2023.
- * Applied Topology Seminar. Oxford University, Britain (Virtual), Nov 2023.
- * Applied Algebraic Topology Research Network (AATRN) Online Seminar. Virtual, Nov 2023.
- * Probability and Statistical Physics Seminar. Chicago University, IL, Oct 2023.
- * Seminario Doctorado, Actividad del Programa de Doctorado "Matematicas". University of Seville, Spain, Sep 2023.
- * Probability and Applications Seminar. Queen Mary University of London, Britain, Sep 2023.
- Computation Persistence Workshop. Purdue University, IN, Sep 2023.
- * Probability Seminar. Purdue University, IN, Sep 2023.
- Geometry and Topology meet Data Analysis and Machine Learning. Northeastern University, MA, Jun 2023.
- ◇ Randomness in Topology and its Applications. The University of Chicago, IL, Mar 2023.
- Finger Lakes Probability Seminar. Binghamton University, NY, Feb 2023.

"Detecting Weak Topological Signals in Noisy Environment".

- Computation Persistence Workshop. Graz University of Technology, Austria (Virtual), Sep 2024.
- * Hot Topics in Data Science. University at Buffalo, NY (Virtual), Feb 2024.
- Joint Statistical Meetings. Toronto, Canada, Aug 2023.
- * Imaging Seminar. The Chinese University of Hong Kong, Hong Kong, Jan 2023.
- Binghamton University Graduate Combinatorics, Algebra and Topology. Binghamton University, NY, Nov 2022.
- 3rd Upstate New York Topology Seminar. Syracuse University, NY, Oct 2022.
- ◇ Algebraic Topology, Methods, Computation and Science 10. Oxford University, Britain, Jun 2022.
- ◇ Bridging Applied and Quantitative Topology. Virtual, May 2022.
- ◇ AATRN Poster Session. Virtual, Oct 2021.

SELECT AWARDS AND HONORS

- Croucher Fellowship for Postdoctoral Research** 2024
Annually, 6 – 10 Hong Kong scholars pursuing overseas postdoctoral research in science are selected.
- Croucher Scholarship for Doctoral Study** 2019
Annually, 9 – 16 Hong Kong scholars pursuing overseas doctoral degrees in science are selected.
- Sir Edward Youde Memorial Fellowship (for Postgraduate Research Students)** 2018
Annually, 3 – 5 Hong Kong fellows are selected among nominees from local institutions.
- Best Teaching Assistant Award at CUHK Math** 2019
Annually, 3 teaching assistants in the Department of Mathematics at CUHK receive this award.

PROFESSIONAL SERVICES

- drafting aims and reviewing drafts of an NIH-R01 and an NSF grant for the lab *Spring 25 – present*
reviewer of *Scientific Reports*, *Electronic Journal of Probability*,
Homology, Homotopy and Applications, and *Mathematical Reviews* *Fall 23 – present*
student representative of the Applied Math Colloquium Committee, Cornell *Fall 23 – Spring 24*
officer of SIAM Student Chapter, Cornell *Fall 22 – Spring 24*

TEACHING EXPERIENCES

- Head Teaching Assistant, Cornell University** *Spring 23*
Facilitated testing accommodations for students with disabilities in “Multivariable Calculus for Engineers”
Drafted and reviewed questions in tests and exams
- Teaching Assistant, Cornell University** *Fall 22*
Led 3 weekly discussion sessions for “Multivariable Calculus for Engineers”
Graded assignments, tests and exams
- Teaching Assistant, CUHK** *Fall 17 – Spring 19*
Designed and led weekly discussion sessions, and graded assignments, tests and exams in 5 courses, ranging from multivariable calculus to complex analysis
Mentored high school students on number theory and cryptography in a summer outreach program by leading thrice-weekly discussion sessions

MENTORSHIP EXPERIENCES

- Mentorship of Graduate Students and Junior Staff** *Fall 24 – present*
Successfully mentored a graduate student through the NSF-GRFP fellowship application process, leading to a successful award in 2026.
Meet biweekly with 3 graduate students to discuss research progress and career development
Provide guidance on mathematical and statistical issues to graduate students and junior staff
- Supervision of Undergraduate Research Assistants** *Fall 21 – Summer 24*
Onboarded and supervised 3 undergraduate research assistants, 2 are now graduate students at University of Wisconsin-Madison and Rice University; 1 is now a financial analyst
Delegated numerical computations and advised on career development
- Mentorship in Undergraduate Directed Reading Program** *Fall 20 – Summer 22*
Mentored 4 students in the Undergraduate Directed Reading Program through weekly meetings
Tailored and discussed reading materials on topics including random graphs and computational topology

ADDITIONAL INFORMATION

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| Natural languages | English, Chinese (Cantonese, Mandarin) |
| Programing | MATLAB, Python, Bash, R, slurm |
| Neuroscience softwares | FSL, XCP-D, Workbench, Nilearn, NiBabel |